Mentimeter

VaR in Practice

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- ✓ *Sit with your group today*

Topics

- Case discussion
- Limitation of VaR
- Expected shortfall
- Variance forecasting

Statistical model for violations

- Violations follow a binomial distribution
  - N= number of observations
  - $p = 1 - X$
- If violations are independent, then the number of violations m follows:

$$\mathrm{m} \sim Binomial(N, p)$$

One-sided Tail test

- H_0: $The\ VaR\ model\ is\ correct$

- P-value: The probability of observing data *at least as extreme* as what we saw, assuming the null hypothesis is true.
  - Small p-value → Very surprising → Reject
  - Calculate right-tail or left-tail probability

VaR Backtesting

Compare with $\mathbb{E}\{\#violations\} = N(1-X)$

If $\#violations < N(1-X)$

- Reject if $\Pr\{m \leq \#violations\} \leq 5\%$.
- $\text{Binomdist}(\#violations, N, 1-X, 1) \leq 5\%$

If $\#violations > N(1-X)$

- Reject if $\Pr\{m \geq \#violations\} \leq 5\%$.
- $1 - \text{Binomdist}(\#violations - 1, N, 1-X, 1) \leq 5\%$

Review Questions: T/F

1. Given that BINOMDIST(2,252,0.01,1)=0.54, we should reject the model with 5% confidence level if we observe 3 violations each year (252 trading days).

1. Given that BINOMDIST(1,252,0.01,1)=0.28, we should NOT reject the model with 5% confidence level if we observe 1 violation each year (252 trading days).

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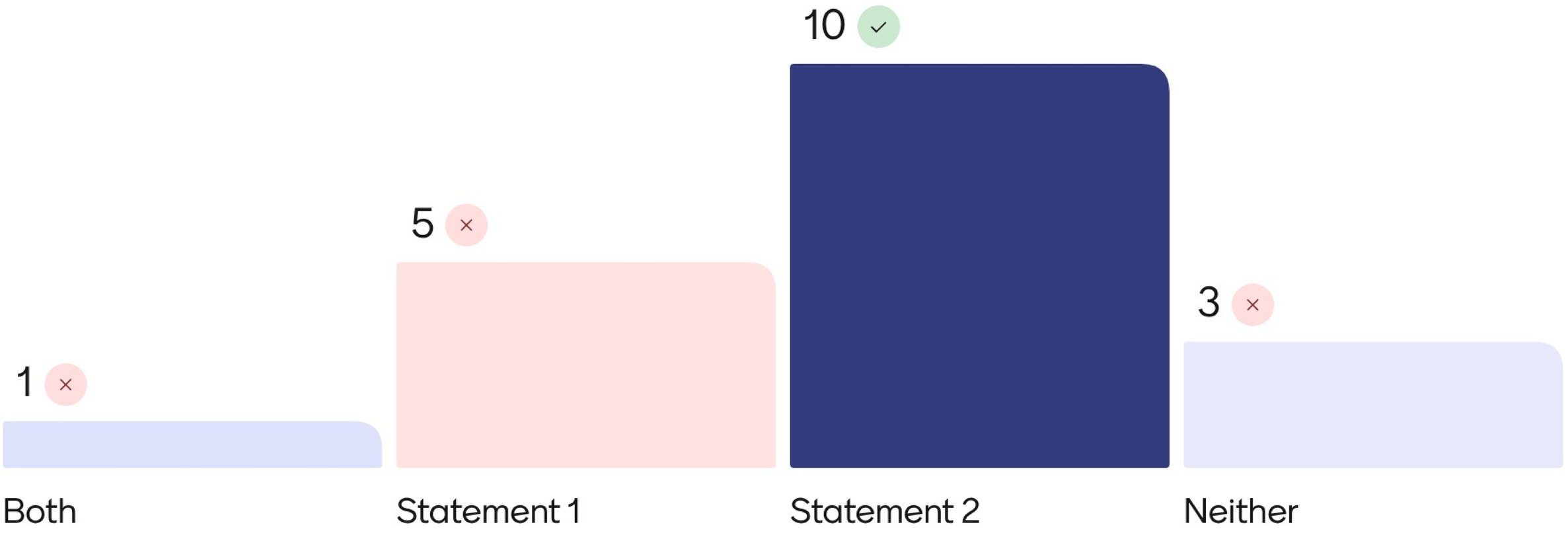
Which statements are corrects?



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Discussion: VaR or not?

- Jasper Wang
- Jianguo Lu
- Charles Pan

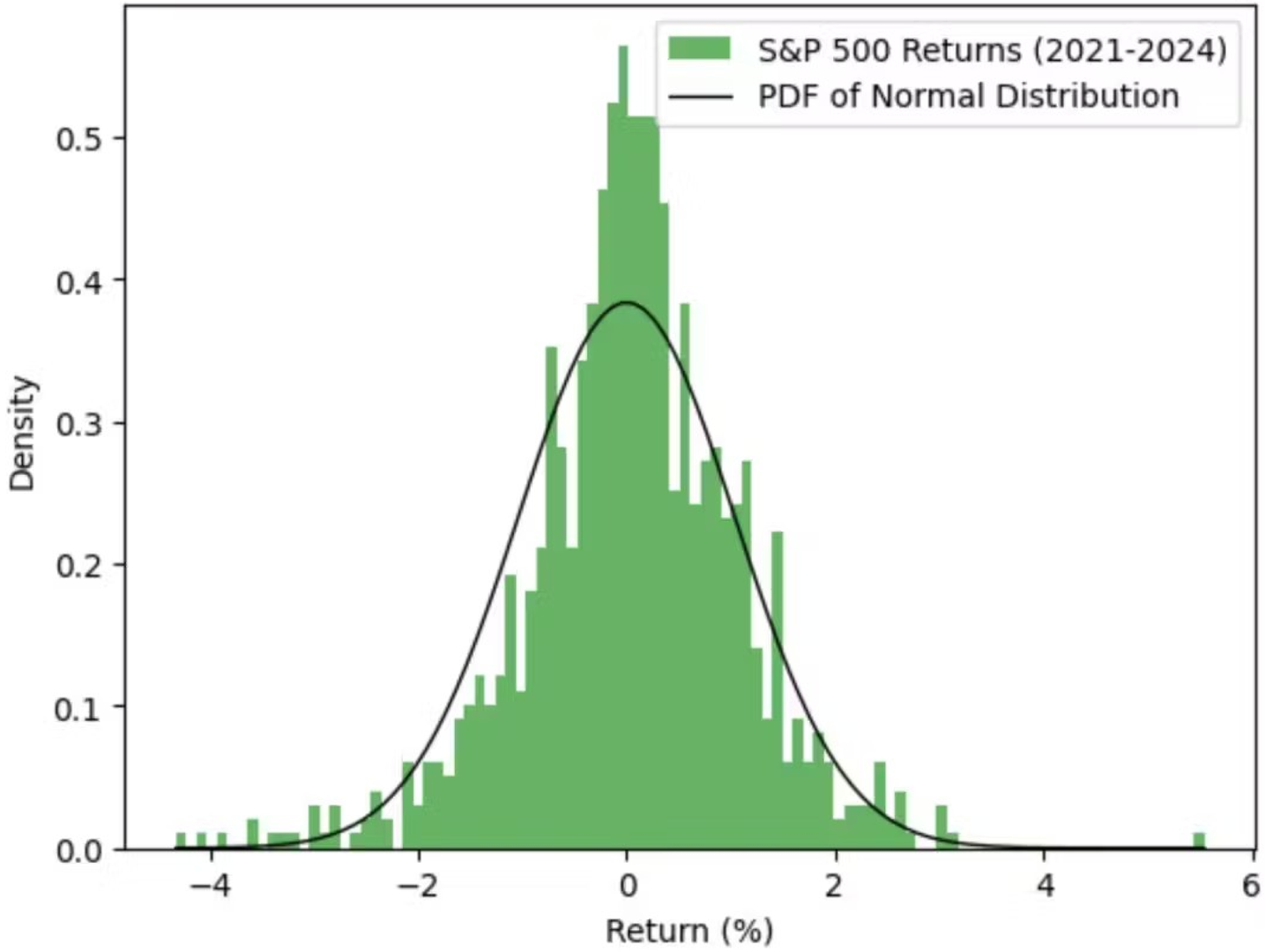
- Your goals:
  - Think about the objectives and priorities of each player
  - Develop a good argument for your positions

Which assumptions are true?

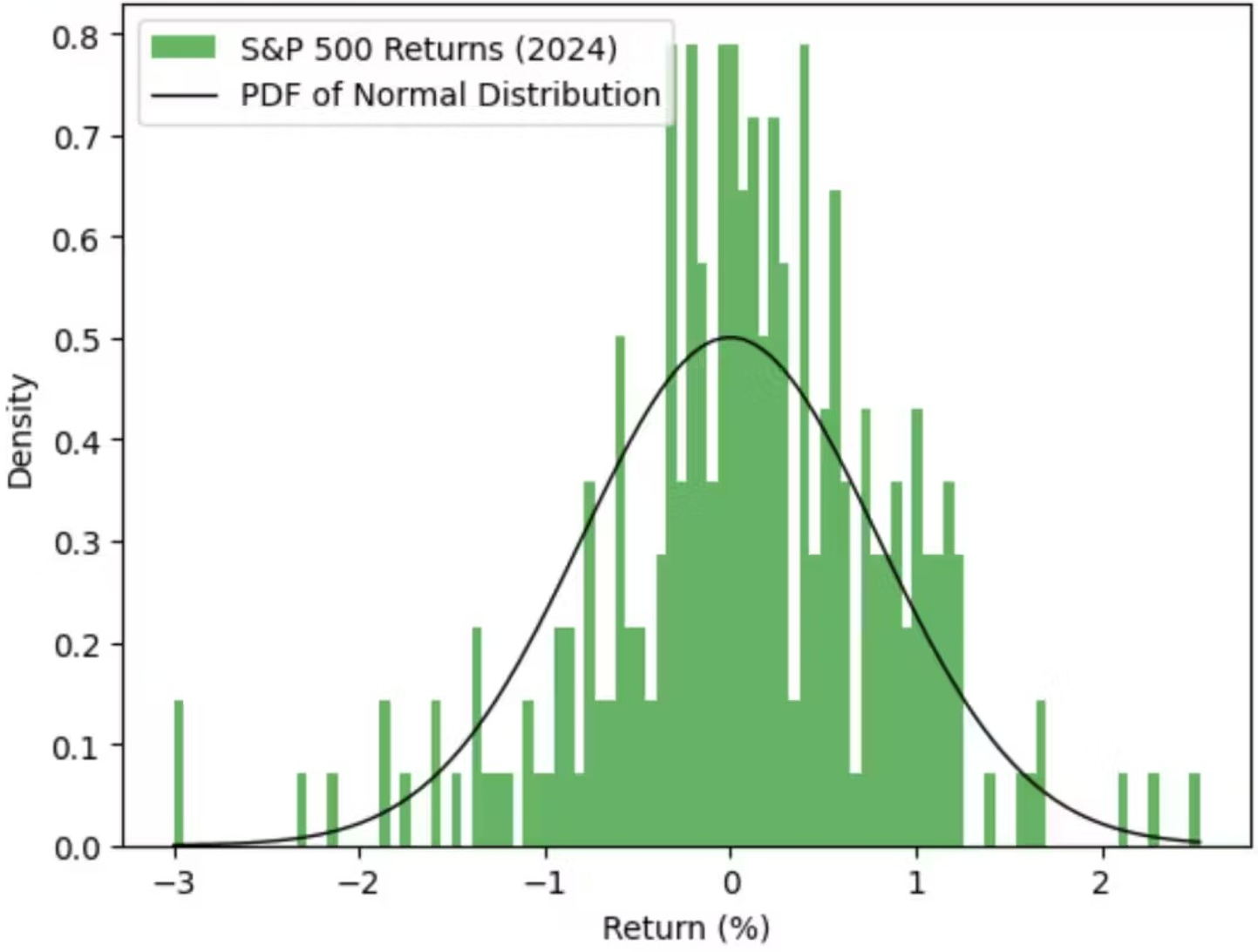
- Returns are normally distributed

- Daily return for each period is i.i.d, where $r \sim N(\mu, \sigma^2)$

Are Daily Returns Normally Distributed?



Are Daily Returns Normally Distributed?



Heavy Tails

- The distribution has heavier tails than the normal distribution
- It is more peaked than the normal distribution
  - This means that small changes and large changes are more likely than the normal distribution would suggest
- Many market variables have this property, known as excess kurtosis,
  - defined as the fourth standardized moments

$$\mathrm{Kurt}[X] = \mathrm{E}\left[\left(\frac{X-\mu}{\sigma}\right)^4\right] = \frac{\mathrm{E}\left[(X-\mu)^4\right]}{\left(\mathrm{E}[(X-\mu)^2]\right)^2} = \frac{\mu_4}{\sigma^4}$$

Problems with VaR

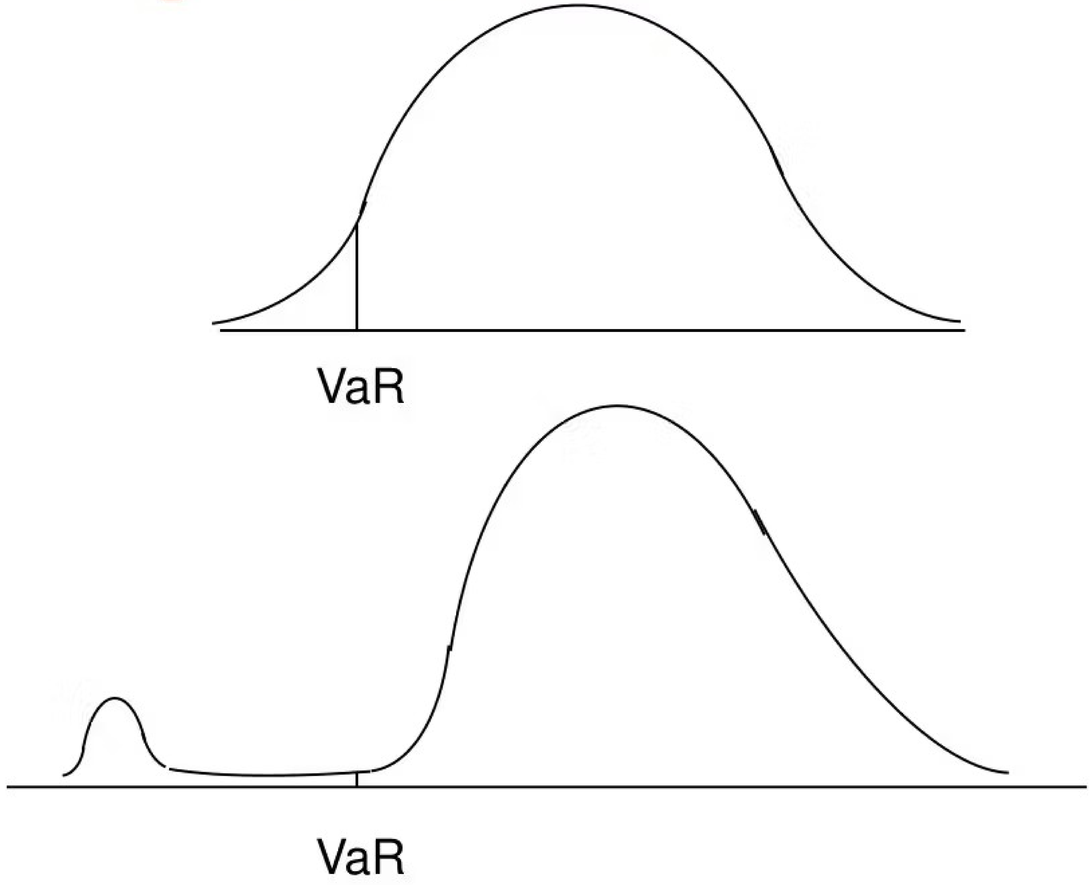
- The normal versus a heavy tail distribution
  - Can have a similar 1% VaR
  - But very different 0.1% VaR

- VaR is only concerned with the number of losses which exceed the VaR but not the magnitude of these losses.

VaR vs. Expected Shortfall

- VaR: “How bad can things get?”
  - The loss level that will not be exceeded with a specified probability
- Expected shortfall (ES): “If things do get bad, what is the expected loss?”
  - The expected loss, given that the loss is greater than the VaR level
- Two portfolios with the same VaR can have very different expected shortfalls
  - Regulators have moved from using VaR to using ES for determining market risk capital

Distributions with the Same VaR but Different Expected Shortfalls



How to get away the risk limits?

- A trader would like to take high risks without violating risk limits

- Possible strategy: writing an out-of-the-money options
  - Collecting option premium most of the time
  - But comes with a big loss with small probability

Normal Distribution Assumption

- When losses are normally distributed with mean μ and standard deviation σ

$$\text{VaR} = \mu + \sigma Y$$

$$\text{ES} = \mu + \sigma \frac{e^{-Y^2/2}}{\sqrt{2\pi}(1-X)}$$

where X is the confidence level and

$$Y = N^{-1}(X)$$

Properties of Coherent Risk Measures

- If one portfolio always produces a worse outcome than another its risk measure should be greater
- If we add an amount of cash E to a portfolio its risk measure should go down by E
- Changing the size of a portfolio by λ should result in the risk measure being multiplied by λ
- The risk measures for two portfolios after they have been merged should be no greater than the sum of their risk measures before they were merged

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Does Expected Shortfall satisfy the 2nd property? Adding E reduces the measure by exactly E

Yes	No
0	0

VaR vs Expected Shortfall

- VaR satisfies the first three conditions but not the fourth one
  - See practice questions

- ES satisfies all four conditions

Which assumptions are true?

- Returns are normally distributed

- Daily return for each period is i.i.d, where $r \sim N(\mu, \sigma^2)$

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Variance forecasting

- variance, as measured by squared returns, exhibits strong autocorrelation

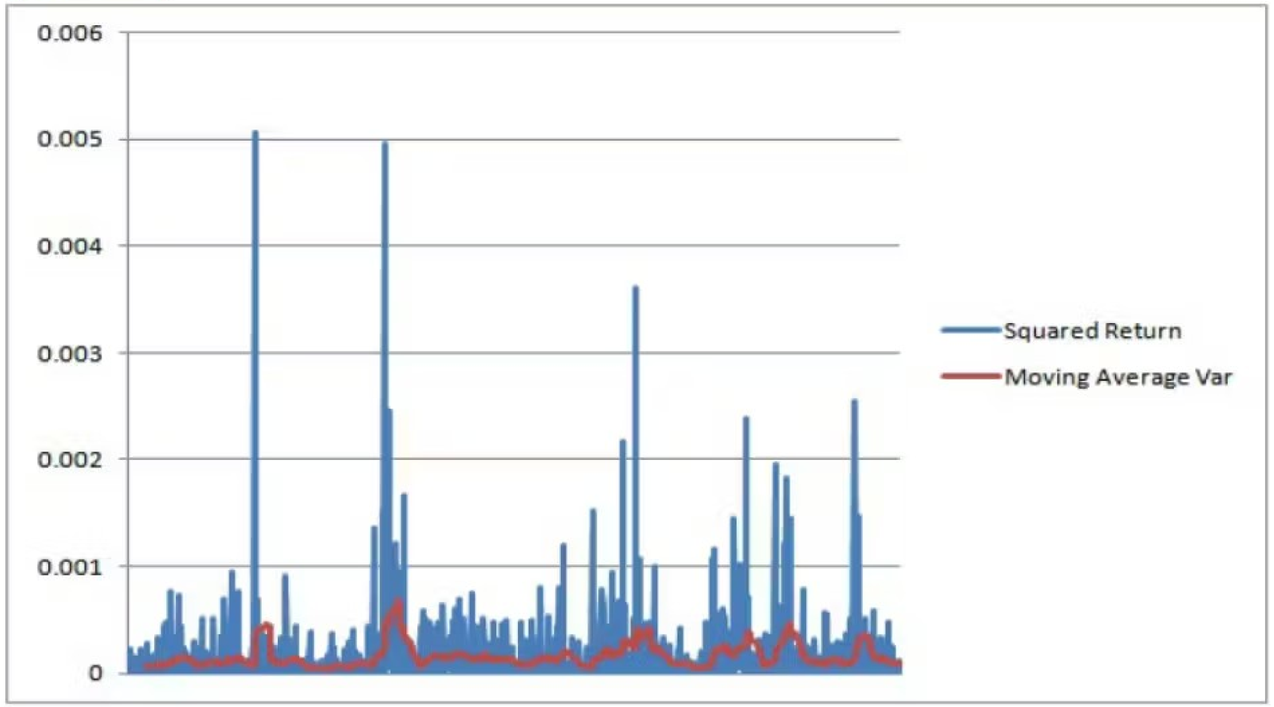
- The easiest way to capture this is by using the simple moving average
  - Some common simplifications: the mean value of $u_i = \ln(S_i/S_{i-1})$, is zero and replace m−1 by m

$$\sigma_n^2 = \frac{1}{m}\sum_{i=1}^{m} u_{n-i}^2$$

S&P 500 Returns

- Volatility tends to cluster: periods of high (low) volatility are likely to be followed by high (low) volatility.

(m=25)



EWMA Model

- In an exponentially weighted moving average model, the weights assigned to the R^2 decline exponentially as we move back through time

$$\sigma_{t+1}^2 = (1-\lambda)\sum_{\tau=1}^{\infty}\lambda^{\tau-1}R_{t+1-\tau}^2, \text{ for } 0 < \lambda < 1$$

It can be rewritten as

$$\sigma_{t+1}^2 = \lambda\sigma_t^2 + (1-\lambda)R_t^2$$

Attractions of EWMA

- Relatively little data needs to be stored
- We need only remember the current estimate of the variance rate and the most recent observation on the market variable
- Tracks volatility changes
- $\lambda = 0.94$ has been found to be a good choice across a wide range of market variables (JP Morgan RiskMetrics)

GARCH Model

- Allows the estimation of variance to vary quickly with recent market moves.
- GARCH stands for Generalized Autoregressive Conditional Heteroskedasticity, which basically means that the variance on one day is a function of the variance on the previous day.
- GARCH assumes that the variance is equal to a constant, plus a portion of the previous day's change in the risk factor, plus a portion of the previous day's estimated variance:

GARCH (1,1)

In GARCH (1,1) we assign some weight to the long-run average variance rate

$$\sigma_n^2 = \gamma V_L + \alpha u_{n-1}^2 + \beta \sigma_{n-1}^2$$

Since weights must sum to 1

$$\gamma + \alpha + \beta = 1$$

Example

- Suppose

$$\sigma_n^2 = 0.000002 + 0.13u_{n-1}^2 + 0.86\sigma_{n-1}^2$$

- The long-run variance rate is 0.0002 so that the long-run volatility per day is 1.4%

Example continued

- Suppose that the current estimate of the volatility is 1.6% per day and the most recent percentage change in the market variable is 1%.
- The new variance rate is

$$0.000002 + 0.13 \times 0.0001 + 0.86 \times 0.000256 = 0.00023336$$

The new volatility is 1.53% per day

Summary: VaR's limitation

- It assumes that the variances and correlations between the market-risk factors are stable.
  - Allowing variance to change over time
  - EMMA, GARCH
- It does not give a good description of extreme losses beyond the 99% level.
  - Expected shortfall can help capture that
  - Allowing for different distributions
- Next: Simulation approach